

Auteur: Ayoub Jaaouan
 Studierichting: Informatica
 Oefeningenreeks 3G nr. 7.5

$$f(x) = (x-1)(x-2)(x-3)$$

Nulpunten:

$$f(x) = 0 \Rightarrow f(1) = (1-1)(1-2)(1-3) = 0 \Rightarrow x = 1$$

$$f(x) = 0 \Rightarrow f(2) = (2-1)(2-2)(2-3) = 0 \Rightarrow x = 2$$

$$f(x) = 0 \Rightarrow f(3) = (3-1)(3-2)(3-3) = 0 \Rightarrow x = 3$$

$$f'(x) = (3x^2 - 12x + 11)$$

Extrema:

$$D = 12$$

$$x_1 = \frac{12 + \sqrt{12}}{6}$$

$$x_2 = \frac{12 - \sqrt{12}}{6}$$

$$f''(x) = (6x - 12)$$

Buigpunten:

$$f''(x) = 0 \Rightarrow f''(2) = (6 \cdot 2 - 12) = 0 \Rightarrow x = 2$$

Tekenonderzoek.:

x	1			$\frac{12 - \sqrt{12}}{6}$	2			$\frac{12 + \sqrt{12}}{6}$	3		
$f(x)$	-	0	+	+	+	0	-	-	-	0	+
$f'(x)$	+	+	+	0	-	-	-	0	+	+	+
$f''(x)$	-	-	-	-	-	0	+	+	+	+	+
$f(x)$	$\nearrow$	$\nearrow$	$\nearrow$	$M\left(\frac{2\sqrt{3}}{9}\right)$	$\searrow$	$\searrow$	$\searrow$	$m\left(\frac{-2\sqrt{3}}{9}\right)$	$\nearrow$	$\nearrow$	$\nearrow$
	$\frown$	$\frown$	$\frown$	$\frown$	$\frown$	0	$\frown$	$\frown$	$\frown$	$\frown$	$\frown$

References

3G-7.5-ayoub.jaaouan.INF.png